1

Solve $\begin{cases} \frac{dy}{dx} = \frac{3x^2+4x+2}{2(y-1)} = f(x,y) \\ y(0) = -1 \end{cases}$

Assumptions: $f(x,y)$ is cont. on $\mathbb{R}$

$f_y(x,y)$ -11- ①

$$\int 2(y-1)\,dy = \int (3x^2+4x+2)\,dx$$

$$y^2 - 2y = x^3 + 2x^2 + 2x + C$$

$$(-1)^2 - 2\times(-1) = C \Rightarrow C = 3$$

$$y^2 - 2y - (x^3+2x^2+2x+3) = 0$$

$$y(x) = \frac{2 \pm \sqrt{4 \pm 4(x^3+2x^2+2x+3)}}{2} = 1 \pm \sqrt{x^3+2x^2+2x+4}$$

⇒ Unique sol. is: $y(x) = 1 - \sqrt{\quad}$

$\Rightarrow (x+2)(x^2+2) > 0$

$\Rightarrow x > -2$

Take $y(0) = 1 \Rightarrow C = 1^2 - 2\times 1 = -1$

$$y^2 - 2y - (x^3+2x^2+2x-1) = 0$$

$$\Rightarrow y(x) = 1 \pm \sqrt{x^3+2x^2+2x}$$ (non-uniqueness) ②

$\rightarrow x(x^2+2x+2) > 0 \Rightarrow x > 0$

Sec. 2.8

$a_n = \frac{1}{n}$

$f(n) = \frac{1}{n}$

① seq. $\to f_n(x) = \frac{x}{n}$, $f_1(x)=x$, $f_2(x)=\frac{x}{2}$, $f_3(x)=\sim$, ...

$f_n : A \to \mathbb{R}$

$x \in A$

$\lim_{n\to\infty} f_n(x) = 0$,

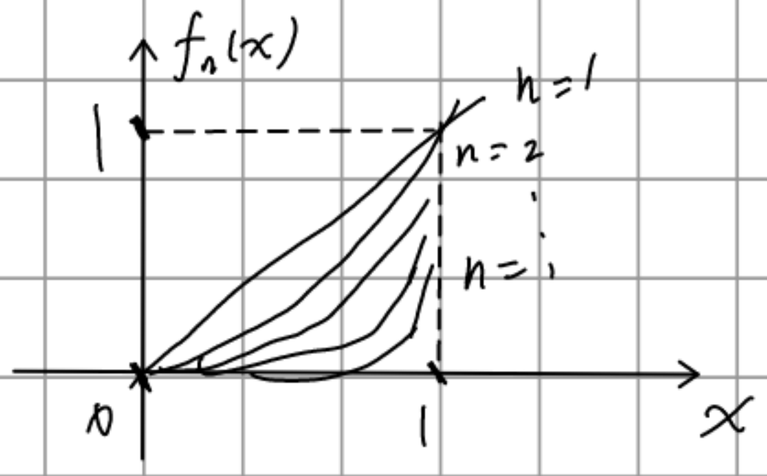
f_n is a seq. of func.'s

② $f_n(x) = x^n$, $\to f_1(x)=x$, $f_2(x)=x^2$, $f_3(x)=\sim$, ...

$x \in [0,1]$

$n = 1, 2, \dots$

$\forall n$, $\begin{cases} f_n(0)=0 \\ f_n(1)=1 \end{cases}$



$\to \lim_{n\to\infty} f_n(x) = \begin{cases} 1, & x=1 \\ 0, & 0 \le x < 1 \end{cases}$

③ $\int_0^1 f_n(x)\,dx = a_n$

If $\lim_{n\to\infty} a_n = L \Rightarrow$ then $\lim_{n\to\infty} \int_0^1 f_n(x)\,dx = L$

?是否: $\int_0^1 \lim_{n\to\infty} f_n(x)\,dx = \lim_{n\to\infty} \int_0^1 f_n(x)\,dx = L$? → 有条件

uniform conv.
pointwise conv.

Let $f_n : A \to \mathbb{R}$ be a seq. of fmc.'s

Def.: $f_n(x)$ conv. to $f(x)$ pointwise

$\Leftrightarrow \forall \varepsilon > 0$ and $x \in A$, $\longrightarrow$ (fixed on x!)

$\exists N \in \mathbb{N}$, s.t. $\forall n > N$, $|f_n(x) - f(x)| < \varepsilon$

Def.: $f_n(x)$ conv. to $f(x)$ uniformly

$\Leftrightarrow \forall \varepsilon > 0$, $\longrightarrow$ (不定 x)

$\exists N \in \mathbb{N}$, s.t. $\forall n > N$ and $x \in A$, $|f_n(x) - f(x)| < \varepsilon$

E.g.: $f_n(x) = \frac{x}{n}$, $f(x) = 0$

Take $N = [\frac{1}{\varepsilon}] + 1$ $\longrightarrow$ $N(\varepsilon)$: 只 dep. on ε ($\leftrightarrow$ uni. conv.)

$|f_n(x) - f(x)| = |\frac{x}{n} - 0|$

$= \frac{x}{n} \leq \frac{1}{n} < \frac{1}{N} = \frac{1}{1 + [\frac{1}{\varepsilon}]} < \varepsilon$

(pw. conv.: $N(\varepsilon, x)$ $\longrightarrow$ dep. on ε and x)

2

Q: If $\lim_{n\to\infty} f_n(x) = f(x)$

then 是否可能有 $\lim_{n\to\infty} \int_a^b f_n(x)\,dx = \int_a^b f(x)\,dx$?

A: X

For $f: [a,b] \to \mathbb{R}$:

Uni. conv. 的另一种 def. :

$f_n(x) \to f(x)$ uniformly

$\iff \lim_{n\to\infty} \max_{x\in[a,b]} |f_n(x) - f(x)| = 0$

Prop.: Let $f_n : A \to \mathbb{R}$

$\{$ f_n is cont. seq. of func.'s

$f_n(x) \to f(x)$ uni. in A

$\Rightarrow f(x)$ is cont.

Proof:

$\forall$ point $x_0 \in A$ (← WTS: $f(x)$ is cont. at $\forall x_0 \in A$)

Let $\varepsilon > 0$ be given

Take $\delta > 0$ s.t. $|x - x_0| < \delta$

Fact:

$f_n \to f$ unif. $\Rightarrow$ we can fix N

s.t. $\forall n > N$, $|f_n(x) - f(x)| < \frac{\varepsilon}{3}$

Fact:

$\forall n$, f_n is cont. at x_0

$\Rightarrow f_{N+1}$ is cont. at x_0

$\Rightarrow (|x - x_0| < \delta \Rightarrow |f_{N+1}(x) - f_{N+1}(x_0)| < \frac{\varepsilon}{3}$

$|f(x) - f(x_0)|$

$= |f(x) - f_n(x) + f_n(x) - f_n(x_0) + f_n(x_0) - f(x_0)|$

$\le |f(x) - f_n(x)| + |f_n(x) - f_n(x_0)| + |f_n(x_0) - f(x_0)|$

↑ 代其上 n 全为 $N+1$

$< \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon$

Take $N = \max$?. $|f_n(x) - f(x)| < \frac{\varepsilon}{3}$

E.g.: ① $f_n(x) = \frac{x}{n}$, $x \in [0, 1]$

$\to\to f_n \to 0$ unif.

($\forall a$ where $|a| < 1$, $\lim_{n\to\infty} a^n = 0$)

② $f_n(x) = x^n$, $x \in [0, 1]$

$f_n(x) \to \begin{cases} 1, & x = 1 \\ 0, & x \in [0, 1) \end{cases} = f(x)$

Prop.: Let $f_n : A \to \mathbb{R}$

$$\begin{cases} f_n \text{ is } \underline{\text{cont.}} \text{ seq. of func.'s} \\ f_n(x) \to f(x) \ \underline{\text{uni.}} \text{ in } A \end{cases}$$

$$\Rightarrow \lim_{n\to\infty} \underbrace{\int_a^b f_n(x)\,dx}_{\text{設為 } a_n} = \underbrace{\int_a^b f(x)\,dx}_{\text{設為 } L}$$

Proof: (WTS: $\lim_{n\to\infty} a_n = L$)

$\forall \underline{\varepsilon > 0}$,

$|a_n - L| = |$ ——— $)$

$= \left| \int_a^b (f_n(x) - f(x))\,dx \right|$

$\leq \int_a^b |\ \downarrow\ |\,dx$

$\leq \max_{x\in[a,b]} |\ \downarrow\ | \cdot \underbrace{\int_a^b dx}_{\hookrightarrow = (b-a)}$

$< \frac{\varepsilon}{b-a} \cdot (b-a) = \varepsilon$

$\left(\lim_{n\to\infty} \left(\max_{x\in[a,b]} |f_n(x) - f(x)| \right) = 0 \right.$

$\left. \Rightarrow \exists N \in \mathbb{N},\ \text{s.t. } \forall n > N,\ \max_{x\in[a,b]} |f_n(x) - f(x)| < \frac{\varepsilon}{b-a} \to 0 \right)$

Solve $\begin{cases} y' = 2t(1+y) \\ y(0) = 0 \end{cases}$

$$y(t) - \underbrace{y(0)}_{(=0)} = \int_0^t y'(s)\,ds = \int_0^t 2s(1+y(s))\,ds$$

$$\Rightarrow \underline{y(t) = \int_0^t 2s(1+y(s))\,ds}$$

$y_0(t) = \underline{y(0) = 0}$

$$y_1(t) = \int_0^t 2s(1+\underline{y_0(s)})\,ds$$

$$= \int_0^t 2s\,ds = t^2$$

$$y_2(t) = \int_0^t 2s\left(1+\underline{y_1(s)}\right)ds$$

$$= \int_0^t 2s(1+s^2)\,ds = t^2 + \frac{t^4}{2}$$

$$y_n(t) = \int_0^t 2s(1+y_{n-1}(s))\,ds$$

$$= t^2 + - - - + - \cdots$$

$(R = \infty)$